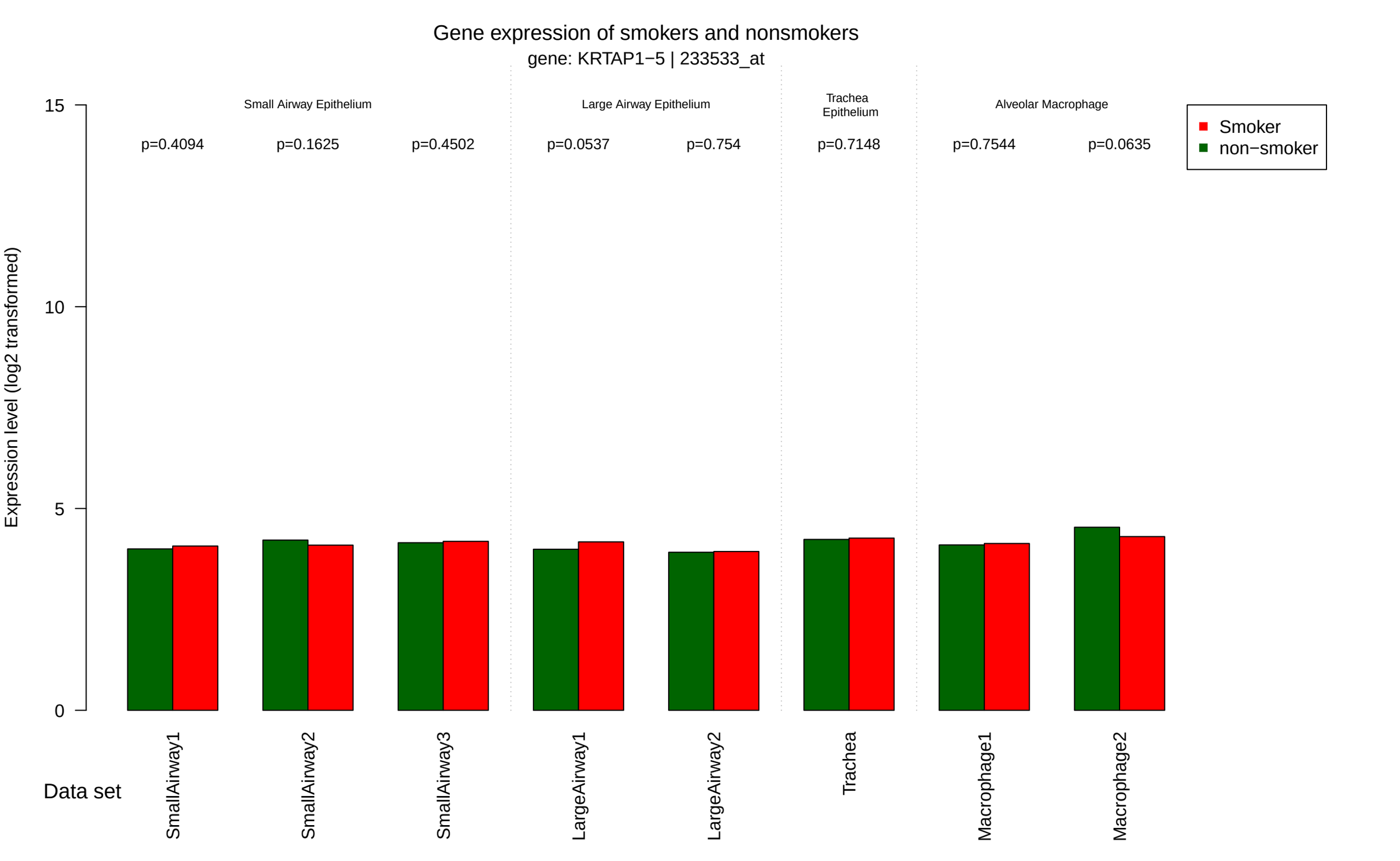
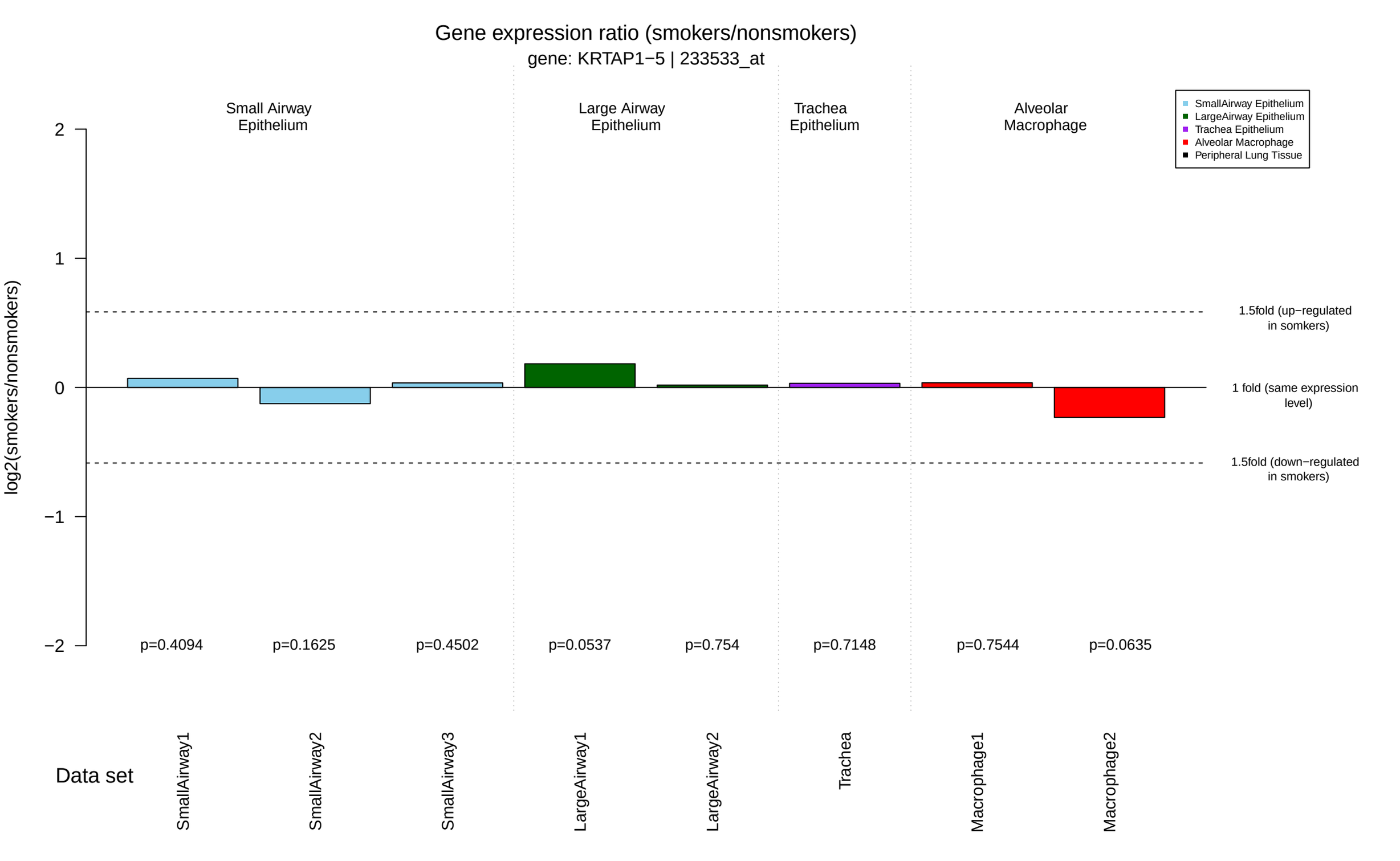
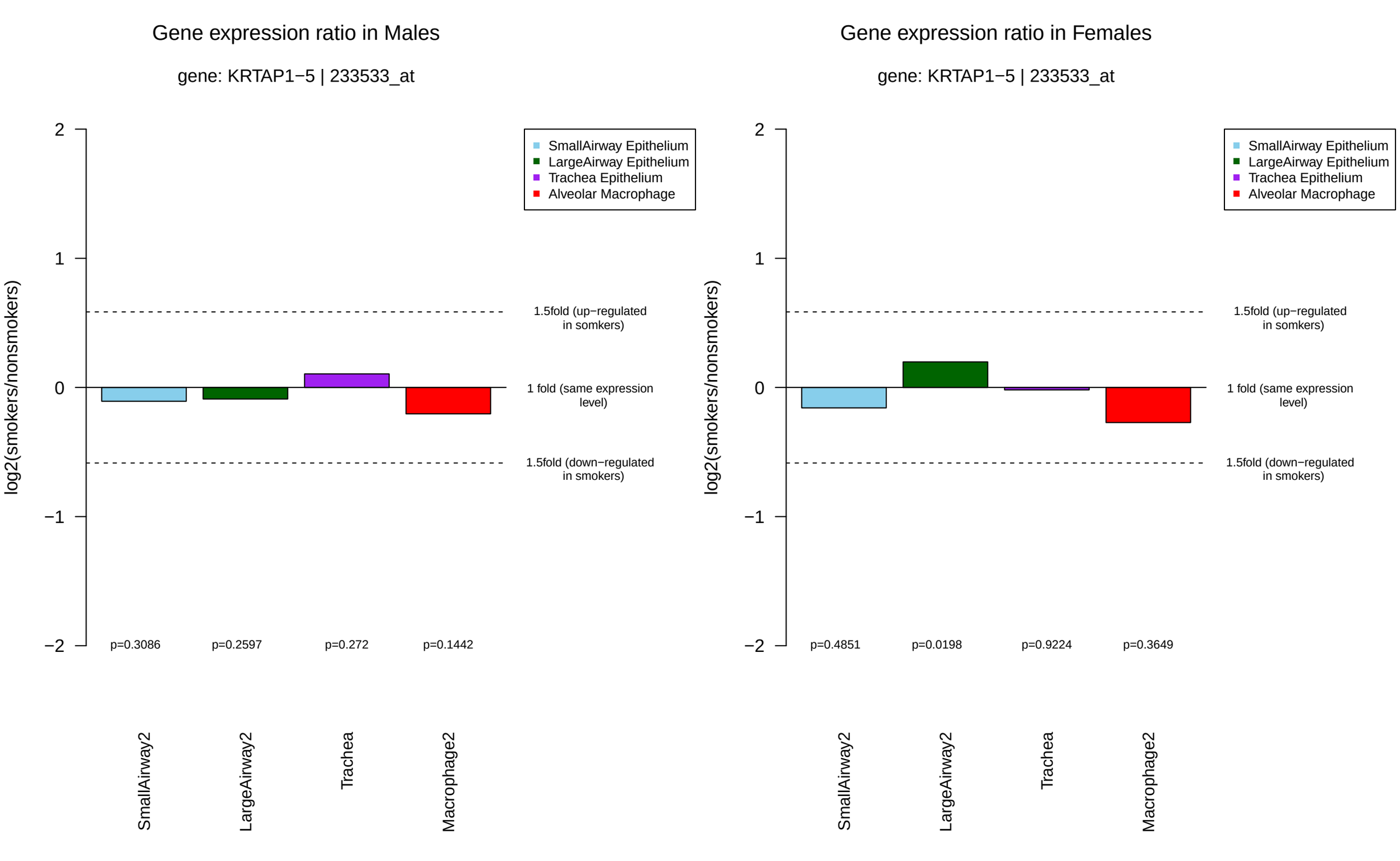
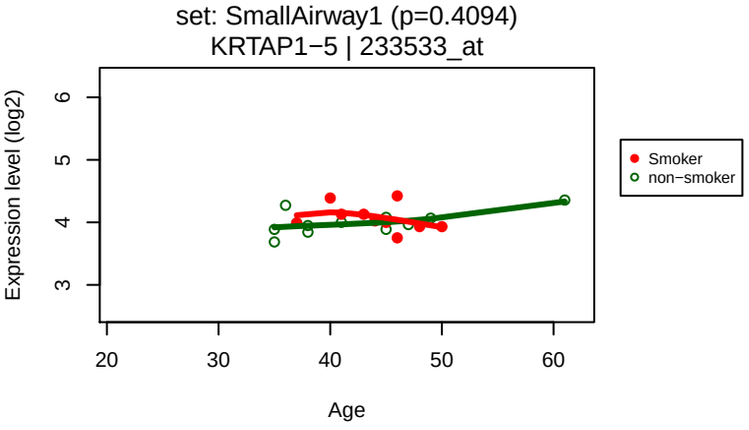
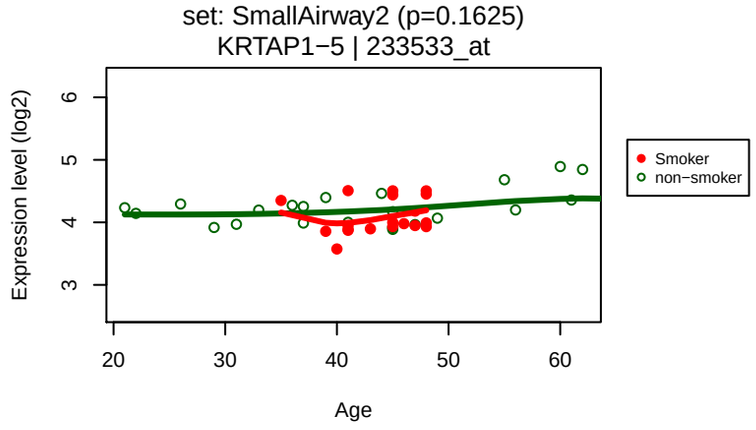
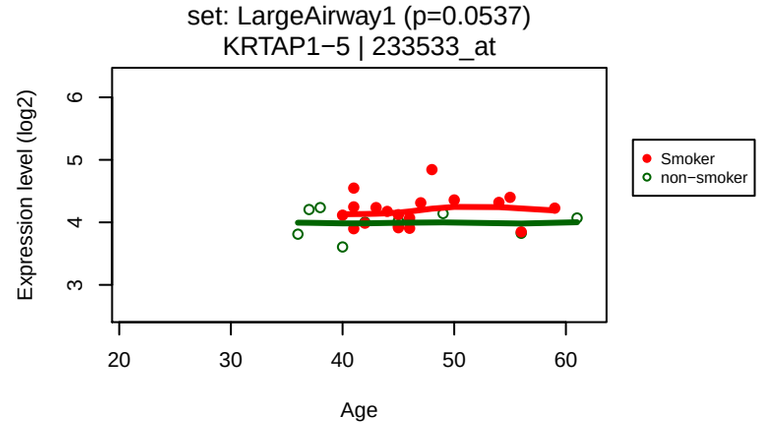

Gene expression of smokers and nonsmokers
gene: KRTAP1−5 | 233533_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4094
p=0.1625
p=0.4502
p=0.0537
p=0.754
p=0.7148
p=0.7544
p=0.0635
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

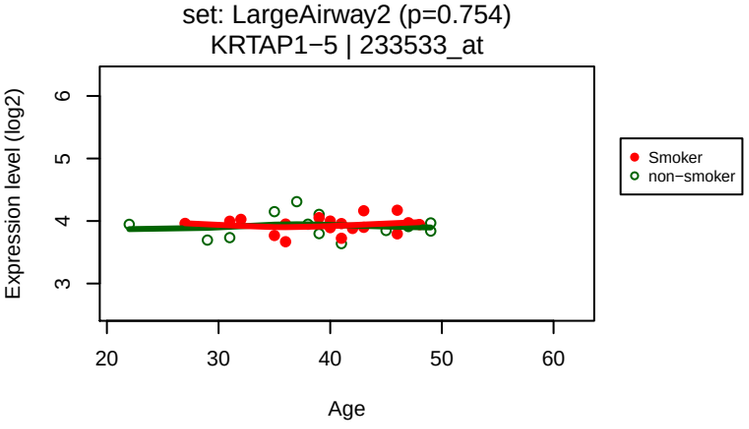

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP1−5 | 233533_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4094
p=0.1625
p=0.4502
p=0.0537
p=0.754
p=0.7148
p=0.7544
p=0.0635
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

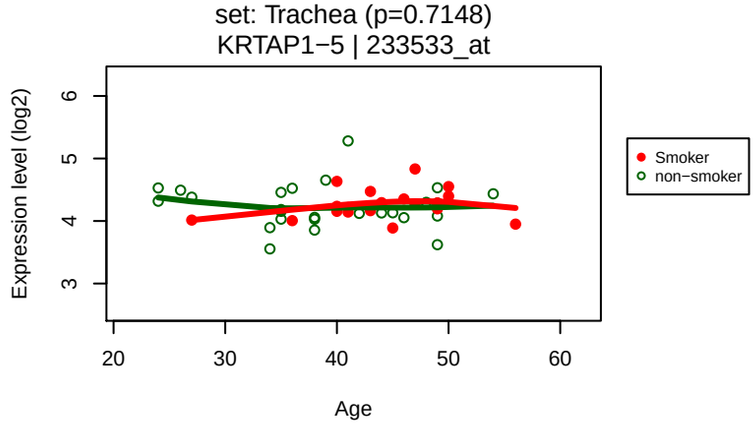

Gene expression ratio in Males
gene: KRTAP1−5 | 233533_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3086
p=0.2597
p=0.272
p=0.1442
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP1−5 | 233533_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4851
p=0.0198
p=0.9224
p=0.3649
SmallAirway2
LargeAirway2
Trachea
Macrophage2

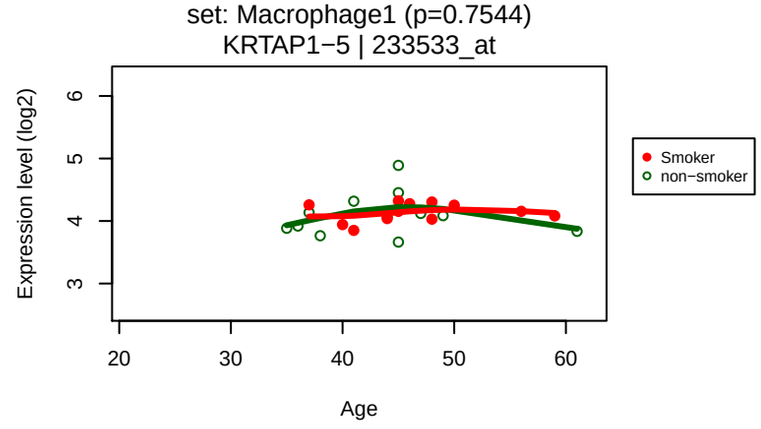

set: SmallAirway1 (p=0.4094)
KRTAP1−5 | 233533_at
Expression level (log2)
Age
Smoker
non−smoker

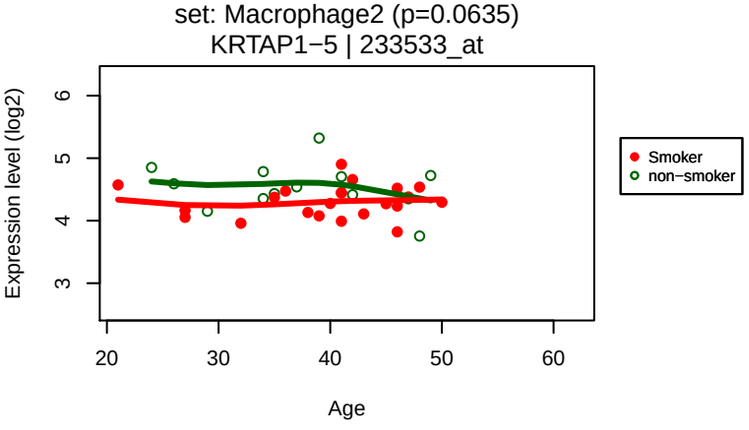

set: SmallAirway2 (p=0.1625)
KRTAP1−5 | 233533_at
Expression level (log2)
Age
Smoker
non−smoker

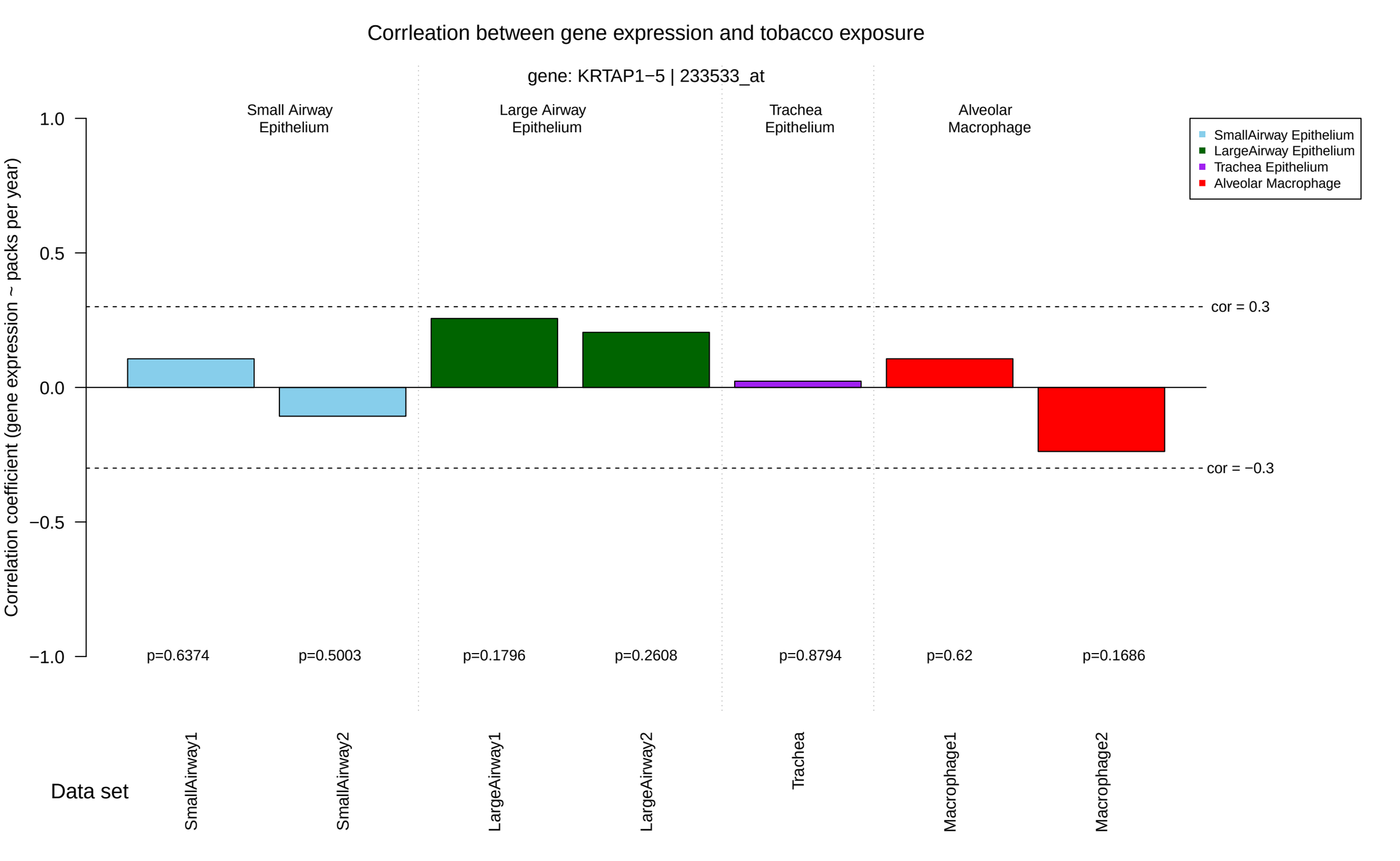

set: LargeAirway1 (p=0.0537)
KRTAP1−5 | 233533_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.754)
KRTAP1−5 | 233533_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7148)
KRTAP1−5 | 233533_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7544)
KRTAP1−5 | 233533_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0635)
KRTAP1−5 | 233533_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP1−5 | 233533_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6374
p=0.5003
p=0.1796
p=0.2608
p=0.8794
p=0.62
p=0.1686
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2